

Multimodal Transformer Serving

Deep Dive into SGLang — Chapter 30

Yin Yang

Foundation Books Project

Where this chapter sits

This is the chapter that turns “pass an image tensor to the model” into a decoder-readiness account the runtime must keep coherent.

- Volume I supplied request state and architecture-name dispatch; a text request is prefill-ready once it has token ids.
- A multimodal request must fetch media, run a processor, execute an encoder, possibly move features across processes, and *then* insert them into reserved token spans.
- One trace carries the chapter: a Qwen/Qwen3-VL-30B-A3B-Instruct request, prompt “What’s in this image?”, with one image input.

The problem is coherence: text positions, feature rows, model metadata, and reserved decoder positions must name *one* request state.

The alignment contract

Family metadata and insertion

Readiness, batching, and placement

Evidence and review

The alignment contract

One coherent prompt, assembled from pieces

Definition (Multimodal request state)

$X_r = (T_r, \mathcal{M}_r, F_r, \alpha_r, \mu_r)$: text tokens, media inputs, feature blocks, feature-alignment map, and model-family metadata for request r .

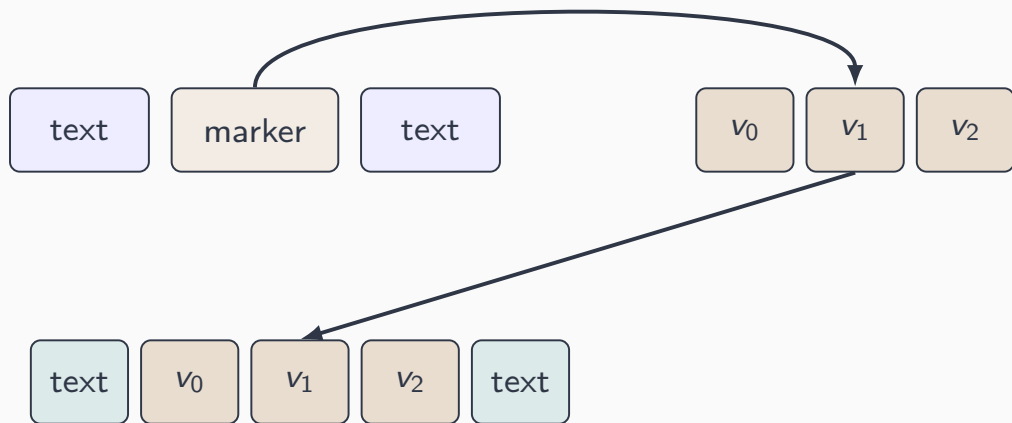
A compact model of the whole path:

$$x_{\text{text}} \oplus z_{\text{vision}} \mapsto y_{\text{text}}.$$

- $\oplus$ is not concatenation: it is token expansion, feature alignment, metadata attachment, and request-state update.
- The decoder-facing prefill cannot begin until $\oplus$ has been paid — so $\oplus$ is part of time to first token.

The chapter's one claim: a feature block is decoder-ready only when its identity, rows, metadata, and reserved positions describe the same prompt.

Feature alignment: markers reserve decoder positions



$\alpha_r(f) = [s_f, e_f)$: reserved positions

Definition (Feature alignment)

Family metadata and insertion

The item record and the insertion rule

Definition (Multimodal item record)

$m = (r, p, \tau, [s, e], F, \mu, \sigma)$: request, part index, modality, reserved span, feature/embedding, family metadata, readiness status.

Definition (Family feature insertion rule)

$l_\theta : (E_T, \{m_i\}) \rightharpoonup E'_T$ — a *partial* function, defined only when every item's token-id, span-length, metadata, and shape requirements hold.

The harpoon is the point: a Qwen-VL item without grid metadata, a Qwen-audio item without a length record, an LLaVA item whose patch-merge span disagrees — each leaves l_θ undefined, so the request must not enter prefill.

M-RoPE: a scalar position becomes an axis vector

Ordinary RoPE treats logical position as one scalar. M-RoPE keeps the rotary principle but carries a $3 \times n$ tensor: temporal, height, width.

$$n_{\text{vis}} = t \begin{bmatrix} h \\ s_h \end{bmatrix} \begin{bmatrix} w \\ s_w \end{bmatrix} = t' h' w'.$$

- A post-merge grid $(1, 24, 24)$ reserves 576 positions; equivalently a raw $(1, 48, 48)$ with 2×2 merge.
- Text rows are the degenerate case: the three coordinates equal the scalar decoder position.
- Corrupt the grid to $1 \times 12 \times 48$ and the same 576 rows get different height/width angles — same shape, different rotary state.

M-RoPE belongs to the insertion contract, not to prompt formatting.

Readiness, batching, and placement

Readiness gates the decoder

Invariant (Multimodal step-alignment)

For every executable batch row, the row index, modality item, feature block, and decoder-position interval must point to one request record.

- **Encode / prefill / decode (EPD)** adds a media-encoding stage before the prefill/decode split.
- A disaggregated request is ready only when *every* expected part arrives once, matches request id and part index, carries no error, and rebuilds the same `mm_inputs` the local path would.
- The first successful encoder tensor proves one part — not readiness.

Ready features attached to the wrong request row are still wrong: the account must reject that state, not charge encoder latency and proceed.

Placement is a tradeoff; the account keeps it honest

$$A_{\text{mm}}(r) = Q_{\text{preprocess}} + E_{\text{encoder}} + X_{\text{transfer}} + M_{\text{alignment}} + D_{\text{decode}}.$$

- **Local:** no network handoff, but the vision tower shares CPU and GPU with the decoder — a ready text prompt can wait behind an image.
- **Disaggregated:** decoder isolation and encoder utilization, but request-part ids, receive ports, transfer buffers, and timeouts.
- Each pre-decoder term is charged to the request id whose spans enter prefill; terms are never aggregated across readiness states.

Placement changes where work runs; it cannot relax the identity, metadata, cache, and family facts that make features decoder-consumable.

Evidence and review

Measure the pipeline, not just the total

Definition (Multimodal TTFT record)

$\Theta_r = (\phi_r, q_r, \ell_r, p_r, e_r, x_r, a_r, d_r, \alpha_r, \sigma_r)$: fixture, queue, load, processor, encoder, transfer, alignment, decode, the alignment record, and ready/timeout/abort/cache status.

- A row with only d_r and total TTFT cannot tell a fast valid path from a fast corrupted feature insertion.
- The review checklist asks five source questions: where markers expanded, which item owns the feature, which metadata is semantic, how remote features became ready, which path was tested.

A benchmark that omits the alignment field may still test text serving — it has not exercised the multimodal contract.

What to carry forward

1. Multimodal serving is **decoder-readiness accounting**, not “text plus an image tensor.”
2. **Processor records** reserve decoder positions; **family metadata** (grid, M-RoPE, length) gives them meaning.
3. l_θ is a **partial function**: a plausible shape with wrong metadata is undefined, not “almost right.”
4. **Readiness** gates admission; identity and part completion, not the first tensor, decide it.
5. **Placement** moves work; the alignment, identity, and metadata obligations stay fixed, and a valid TTFT row proves them.

Next: from the alignment contract to media preprocessing
— how one prompt's markers, processor output, and grid
metadata first become reserved decoder token spans.